

# CODE-LAYOUT, READABILITY AND REUSABILITY

| Field         | Details                   |
|---------------|---------------------------|
| Date          | 29 September 2026         |
| Team ID       | 6ab7959e4c4a9ef0de66bbbee |
| Project Name  | AI FitTrack API           |
| Maximum Marks | 5 Marks                   |

| S.No | Code Quality Parameter    | Followed         | Remarks                                                                                               |
|------|---------------------------|------------------|-------------------------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | To verify        | Source code was not supplied for direct inspection.                                                   |
| 2    | Proper File Structure     | Yes (documented) | Documentation describes separate models, controllers, routes, middleware, services and configuration. |
| 3    | Meaningful Variable Names | To verify        | Requires source-code inspection.                                                                      |
| 4    | Function / Method Names   | To verify        | Requires source-code inspection.                                                                      |
| 5    | Code Comments             | To verify        | Inline source comments cannot be confirmed.                                                           |
| 6    | Modular Design            | Yes (documented) | MVC/layered architecture separates routing, business logic, database, middleware and AI services.     |
| 7    | No Redundant Code         | To verify        | Requires source-code inspection.                                                                      |
| 8    | Error Handling            | Yes (documented) | Centralized error handling and validation are described.                                              |

## Reusable Components / Modules

| S.No | Component / Module Name    | Language / Technology  | Where Reused                                | Reusability Level |
|------|----------------------------|------------------------|---------------------------------------------|-------------------|
| 1    | Authentication Middleware  | JavaScript / Express   | Protected APIs                              | High              |
| 2    | Gemini AI Service          | JavaScript / Node.js   | AI recommendation and fitness insight flows | High              |
| 3    | JWT Service                | JavaScript / Node.js   | Authentication token creation/verification  | High              |
| 4    | Password Service           | JavaScript / bcrypt.js | Registration and login                      | High              |
| 5    | Database Connection Module | Node.js / Mongoose     | Application startup and database access     | High              |

## Overall Code Quality Assessment

| Aspect                   | Rating (1-5)                             | Comments                                                                                  |
|--------------------------|------------------------------------------|-------------------------------------------------------------------------------------------|
| Code Layout & Structure  | 4 (documented architecture)              | Clear modular MVC/layered structure is described; source verification is needed.          |
| Readability              | To verify                                | Cannot fairly rate source readability without the code.                                   |
| Reusability              | 4 (documented)                           | Dedicated middleware and service layers are described as reusable.                        |
| Documentation / Comments | 4 (project documentation)                | The supplied project documentation is detailed; inline source comments were not verified. |
| Overall Score            | To be finalized after source-code review | Final 1–5 score should be assigned after inspecting/running the repository.               |